Supplementary File 3.
Summary of characteristics, measurements, and findings of studies examining food and drink consumption while watching TV.

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Observational/free-living studies							
Gemming 2015 [39] Quality = good	Cross-sectional New Zealand	Recruited from the community and university (convenience)	Healthy adults in Auckland, n=40 (50%) Mean ages 35 (SD=17) and 28 (SD=7) for men and women, respectively	Instances of television time (only while eating) as identified during processing of SenseCam (wearable camera) recordings across 3 days	NA	Energy intake (kJ), energy density (kJ/g), and duration (min) of meals and snacks across contexts (including during TV) measured by 24-hour multiple pass dietary recall on days that wearable camera was worn	Snacks at home when viewing television were higher in energy intake (+514 kJ (47, 1077)) compared to snacks when no television was viewed. Meals at home while watching TV were not significantly longer (3.1 min (-0.6, 6.7)) than meals not viewing screens, and were not significantly higher in energy (543kJ (-32, 1120)).
Ghosh Roy 2018 [40] Quality = good	Cross-sectional USA *African American Women's Daily Life Study*	Recruited from community and university (convenience)	African American women aged 25-65 in Chicago, n=79 (100%) Mean age 44.6 (SD=10.6)	In each instance of reported snack or sweetened beverage consumption (see 'outcome measurement'), participants were asked what they were doing while eating or drinking; watching television was an option	Education, per capita income, age, auto ownership, BMI, weekday or weekend day measurement, hunger status, fast food restaurant and convenience store density in home neighborhood	Self-reported consumption of ≥1 snack food item (including French fries, salty snacks, cookies or sweetened baked goods, chocolate/candy, ice cream/frozen dessert) or ≥1 sweetened beverage in response to smartphone prompts which came 5 times per day across one week (ecological momentary assessment)	Consumption of snack foods was significantly more likely to occur while watching TV versus while doing nothing else (OR 1.8 (95%CI 1.2, 2.7)); no such association was seen for sugar-sweetened beverage consumption (OR 0.9 (95%CI 0.5, 1.6)).
Laska 2011 [41] Quality = fair	Cross-sectional USA	Recruited from community and	Young adults aged 18-23 from four groups (attending	All instances of eating and drinking were logged on a	N/A	Proportion of food consumption while watching TV compared	A larger proportion of calorically sweetened beverages (24% vs 15%) and

		university (convenience)	university and living on campus; attending university and living independently off-campus; attending university and living with parents; not attending university and living independently), n=48 (12 from each group), 56% female Mean age approximately 21	PDA over a 7-day period, and followed up with contextual questions, including whether they were watching TV during that instance of eating		to while not doing anything else, separately by food group: calorically sweetened beverages, water, coffee/tea drinks, milk, cookies and sweetened baked goods, candy/gummy fruit snacks, fruits and vegetables (fresh or 'other'), frozen desserts, non-milk dairy products, entrees, cereals/grains, and fried side dishes	smaller proportions of water (12% vs 17%), fruits and vegetables (26% vs 36%), and cereals/grains (25% vs 31%) were consumed while watching TV compared to while not doing anything else, respectively ($p<0.05$ for all). There were no significant differences among the other food groups listed in the previous column.
Stroebele 2004 [42] Quality = fair	Cross-sectional USA	Recruited from university for course credit	University students, n=78 (82%) Mean age 22 (0.9)	Television time was recorded on a diary in 15-minute intervals. Instances of eating while watching TV were recorded on food diaries (see 'outcome measurement' column)	N/A	Meal sizes, caloric intake, and meal frequency, ascertained from a 7-day food diary in which participants recorded the volume of all food and drinks consumed, with a specification concerning whether the TV was on in each instance	Meal sizes did not differ with and without TV (data not shown). There was no correlation between TV time (h/day) and total caloric intake or meal frequency (data not shown).
Experimental/laboratory studies							
Study, quality	**Study design, country**	**Target sample Sample size (% female) Mean (SD) age or age range**	**Study aim and experimental protocol**		**Main outcome measurement**	**Findings**	
Anschutz 2011 [33] Quality = fair	Between subjects Netherlands	Non-overweight university students, n=82 (50%) Mean age (SD) for men and women was 20.9 (2.5) and	Aim: to examine whether food commercial exposure would increase concurrent snack food intake among young adults Protocol: participants were randomized to 'food commercial' condition (30-minute nature TV clip interrupted by 3 food and 5 non-food commercials) or 'neutral		Food intake (kcal) based on weight of crisps and M&Ms consumed during the session Amount of food provided not specified	There was no effect of commercial condition on food intake (b=-0.52, SE=43.12, $p>0.05$) and the addition of commercial condition to the model had a negligible effect (change in R^2 <0.001); the interaction between sex and commercial condition was significant (women ate more when exposed to food commercials compared to neutral, while men ate more when exposed to neutral compared to food commercials)	

		20.4 (1.5), respectively	commercial' condition (30-minute nature TV clip with 8 non-food commercials)		
Bellisle 2004 [20] Quality = fair	Within subjects France	Women aged 18-50 with BMI between 18.5 and 24.9, n=48 (100%) Mean age 29.9 (1.4), range 18-50	Aim: To assess the impact of two environmental stimuli (TV and audio story) on meal intake Protocol: Participants ate a meal in four conditions each lasting minimum of 30 minutes and spaced one week apart: control (alone), while watching TV (no references to food or eating), while listening to a recorded detective story, and last meal (alone); order of TV and auditory conditions was randomized	Food intake (kJ) of meal (shepherd's pie and fruit sherbet) ~4850kJ (1160kcal) available	Compared to the baseline control condition (1751 (SE=84) kJ), meal size was larger in the TV (2023 (SE=84) kJ) and audio recording (2044 (SE=84) kJ) conditions ($p<0.001$). Meal size did not differ between the TV and audio recording conditions ($p>0.05$).
Bellisle 2009 [23] Quality = fair	Within subjects France	Healthy women of normal weight, half with high and half with low dietary restraint, total n=40 (100%) Overall mean age or range not reported	Aim: To assess the effects of dietary restraint and environmental factors on meal intake Protocol: Participants ate a meal in five conditions (randomized order) each lasting a minimum of 30 minutes and spaced one week apart: alone in a quiet room, in groups of 3, alone with TV (no food cues), alone with TV (including food advertisements), alone while listening to audio detective story	Food intake (g, kJ) of meal (shepherd's pie and fruit sherbet) ~6000kJ (1430kcal) available	Significantly less food (g, kJ) was consumed in the group meal condition compared to the others ($p<0.02$) in the pooled sample. Consumption was not different between the other conditions (data not shown).
Blass 2006 [29] Quality = fair	Within subjects USA	Undergraduate students, n=20 (75%) Mean age or range not reported	Aim: to test whether more was eaten when watching TV compared to not Protocol: participants were assigned to either pizza or mac and cheese (assignment protocol not stated) and either watched TV for 30 minutes (program of choice with original commercials included) or listened to *Rachmaninoff's Second Symphony* for 30 minutes; the experiment was repeated a week later for the other condition (counterbalanced order)	Food intake (kcal) of pizza or macaroni and cheese 12-inch pizza or family-size bowl (~900g) of macaroni and cheese available	More calories were consumed with the TV on (793.7kcal) than with TV off (538.2 kcal), $p<0.001$. There was no interaction effect between food type and condition.
Bodenlos 2013 [38] Quality = fair	Between subjects USA	Undergraduate students, n=80 (73%) Mean age 19.5 (1.0), range 18-22	Aim: to test whether exposure to a cooking show affected caloric intake Protocol: Participants were randomized to watch either a cooking show or Planet Earth for 10 minutes, followed by a 'taste test' of cheese curls, chocolate covered candies, and carrots	Intake (kcal) of cheese curls, chocolate covered candies, and carrots 800kcal in total available	No significant difference in overall calories consumed between the two conditions ($F(1,74)=3.32$, $p=0.07$), $d=0.38$ Those in the cooking show group consumed significantly more calories from chocolate covered candies than those in the nature group ($F(1,74)=3.90$, $p=0.05$), $d=0.51$

Boyland 2017 [34]	Within subjects UK	Females recruited from university (staff and students) and surrounding area, n=55 (100%) Mean age 32.4 (9.8), range 20-62	Aim: to examine consumption responses to televised food commercials in overweight and lean adult females Protocol: Participants attended two sessions in a counterbalanced order: TV show including commercials for high-calorie palatable foods and TV show including non-food commercials. Following the show, participants were given pizza to eat.	Intake (kcal) of pizza 2056kcal of pizza available	No main effect of condition or weight status on pizza intake or interaction between weight status and condition (p values >0.28).
Braude 2014 [21] Quality = fair	Within (and between) subjects Australia	Female undergraduates, n=62 (100%) Mean age 19.6 (2.2), range 18-29	Aim: To test whether TV affected sensory specific satiety Protocol: Participants were randomized to a single- or variety-foods group; both groups took part in a TV (*Friends,* no commercials) and no-TV condition (counterbalanced order) lasting 20 minutes and spaced one week apart. In the single-foods group, participants chose their preference between Maltesers, Skittles, almonds, or salted potato chips and were given 80g of it. Participants in the variety-foods group were given a 20g bowl of each food (totaling 80g).	Food intake (kJ, number of items) Variety condition totaled ~1600kcal available; single food condition depended on which food was picked	Participants consumed significantly more energy (z=2.37, p<0.025) and more items of food (z=2.07, p<0.05) in the TV condition compared to the no-TV condition.
Chapman 2014 [36] Quality = fair	Within subjects Sweden	Healthy female women, n=18 (100%) Mean age 22 (1.3)	Aim: to explore how the content of TV programs differentially impacts simultaneous eating behavior Protocol: Participants took part in three conditions (counterbalanced order) each lasting 30 minutes and spaced one week apart: watching an engaging TV program (comedy), a boring program (art lecture) or reading a non-engaging text about insects. Bowls of M&Ms and grapes were available in each condition.	Food intake (grams, kcal) 250 grams (~1200kcal) of M&Ms and 300 grams (~200kcal) of grapes were available	Significantly more food (grams) was consumed during the boring program (125g) compared to the engaging program (82g, p<0.01), but no difference in kcal (data not shown). No differences in grams or kcal consumption were reported between the engaging program vs text condition (109.3g vs 81.9g, p=0.05) or the boring program vs text conditions (+15.5 grams, p=0.26) (kcal data not shown).
Hetherington 2006 [22] Quality = fair	Within subjects UK	University staff and students, n=37 (43%) Mean age 28.3 (1.8), range 18 to 54	Aim: To test whether the social facilitation of eating occurs as a function of distraction (by comparing consumption in social contexts with TV and control conditions) Protocol: Participants took part in four conditions in a counterbalanced order at least 3 days apart: eating alone, eating with TV (game show, duration not fixed), eating with strangers, and eating with friends. The	Food intake (kJ) A buffet-style meal (13,743kJ/3283kcal) with 9 different food items was available	Participants consumed significantly more when eating in front of the TV (4350 (SE=252) kJ) and when eating with friends (4565) SE=272) kJ) compared to the control condition (3861 (SE=200) kJ). No differences were found between eating with strangers and any other condition.

			same buffet meal was available in all conditions.		
Higgs 2009 [31] Quality = fair	Within subjects UK	Female undergraduates of normal weight, n=16 (100%) Mean age 19 (1)	Aim: to examine whether watching TV during lunch would increase afternoon snack intake due to impaired memory Protocol: Participants had a standardized lunch either in quiet solitude (control) or with the TV on (comedy show with no references to food). Both conditions lasted 10 minutes and all participants ate the entire lunch in that time period. At least 2.5 hours later on the same day, participants were given three kinds of cookies broken into small pieces to taste test.	Amount (g) of cookies consumed Three kinds of cookies (40g of each) were available, totaling ~600kcal available	Participants consumed significantly more grams of cookies after the lunch-with-TV condition compared to after the lunch-without-TV condition ($F(1,15)=12.6$, $p<0.01$)
Lyons 2012 [30] Quality = fair	Between subjects USA	Adults aged 18-35, n=120 (50%) Mean age 24.1 (4.4)	Aim: to compare differences in energy intake among 2 sedentary screen behaviors (TV and video games) and one potentially active screen behavior (motion-controlled video game). Protocol: Participants were randomized to a TV, video game (PS3), or motion-controlled video game (Wii or Xbox 360) group. Each condition lasted 1 hour. During the condition, snacks and beverages were available for consumption.	Amount (kcal) of snacks and soda consumed Snack foods included M&Ms, baked Lay's, Doritos, and trail mix (nuts with dried fruit) (amounts provided not specified); soda included cans (3 of each) of Coca-Cola, Diet Coke, and Mountain Dew.	Energy consumption between the three groups was not significantly different ($p=0.07$). Mean intake during TV was 716 (407) kcal; during video games was 747 (540) kcal; and during motion-controlled video game was 553 (498) kcal.
Martin 2009 [25] Quality = good	Within subjects USA	Healthy adults aged 18 to 54 (BMI 20-35), n=48 (54%) Mean age 31.9 (SE=1.5)	Aim: to test the effect of TV viewing with and without ads on energy intake compared to a control and a reading condition Protocol: participants completed four conditions (each lasting 32 minutes) in a random order on two test days (experimental lunch and dinner after consuming a standardized breakfast): control, reading, TV without ads, TV with ads (half food-related, half not). The same buffet meal was available for all conditions.	Food intake (kcal) Buffet-style meals were available including 19 different items (totaling 4921kcal)	Energy intake did not vary by experimental condition ($F(3, 131)=0.30$, $p=0.81$); partial $\eta^2 \leq 0.01$
Mathur 2015 [37] Quality = fair	Within subjects Australia	Female undergraduates, n=45 (100%)	Aim: to test whether variability in engagingness of TV affects food intake	Food intake (g, kJ) of their first and	Significantly more snack food was consumed in the same condition (mean 76.2g, SD=36.2) than the different

		Mean age 19.5 (SD=2.2), range 18-29	Protocol: Participants completed two conditions in a randomized order one week apart: watching the same episode of *Friends* twice back-to-back ('same'), or 'watching two different episodes back-to-back ('different'), with each episode lasting 20 minutes and without advertisements). The first episode was watched without food; for the second episode, regardless of condition, they were given 80g each of their preferred two snacks (given the choices of almonds, Pringles, and M&Ms).	second preferred snacks	condition (mean 66.7g, SD 37.3, z=1.990 (*one-tailed), p<0.05).
Mittal 2010, experiment 1 [24] Quality = fair	Between subjects Australia	Females from the university with BMI between 18 and 25, n=32 (100%) Mean ages 20.8 (SD 3.8) and 20.3 (SD 3.9) in the two groups	Aim: to assess whether snacking with concurrent TV would result in eating more at a test meal later on Protocol: Participants were assigned to either a snack-with-TV or snack-without-TV condition, both lasting 20 minutes (TV program was choice between *Seinfeld* and *Friends*). In both conditions, they consumed as much as they wanted of their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) of the snack in the TV versus no-TV condition 2080kJ (~500kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions 2470kJ (~600kcal) offered (sandwiches, biscuits, crackers and dip)	Energy consumption of the snack did not differ between the snack-with-TV (1855.9 (264.2)kJ) and snack-without-TV (1667.7 (362.0) kJ) conditions (test statistics not shown). Energy consumption of the test meal was significantly higher among those who had watched TV with their snack (1584.6 (516.4) kJ vs 1354.9 (335.6) kJ), p<0.05 (*one-tailed)
Mittal 2010, experiment 2 [24] Quality = fair	Between subjects Australia	Females from the university, n=84 (100%) Overall mean age not reported but around 21 years	Aim: to determine whether the content of TV shows watched while snacking affect recall accuracy of food eaten during the show and intake on a later TV-free test meal Protocol: Participants were assigned to one of four conditions, each lasting 20 minutes: snack-without-TV, snack-with-boring-TV, snack-with-funny-TV, or snack-with-sad-TV. They consumed their snack (potato chips, chocolate balls, and coke or orange juice). Then, after 45 minutes of psychological tests, they were given lunch (test meal).	Food intake (kJ) between the snack-TV conditions 2411kJ (575kcal) offered (chocolate balls, Pringles, Coke/orange juice) Food intake (kJ) of the test meal following the snack conditions	There was no difference in the energy consumption of the snack across the four groups (test statistics not shown). Mean snack consumptions (kJ) were 2308.3 (262.4) for boring, 2291.6 (223.5) for sad, 2250.2 (289.7) for funny, and 2194.9 (322.0) for control. Energy consumption of the test meal was significantly higher in all three TV conditions compared to the no-TV condition (post-hoc pairwise p<0.02 in all three cases), but consumption did not differ between the three TV groups (test statistics not shown). Mean test meal consumptions

				4021kJ (960kcal) offered (sandwiches, crackers and dip, ice cream and chocolate, Coke/orange juice)	(kJ) were 2507.0 (438.2) for boring, 2842.0 (452.4) for sad, 2637.6 (540.3) for funny, and 2147.9 (527.2) for control.
Moray 2007 [28] Quality = fair	Within subjects USA	University students, n=20 (50%) Mean age 20.8 (range 18 to 23)	Aim: to determine whether TV promotes increased food consumption because it impairs one's ability to accurately estimate food intake Protocol: participants completed two conditions in counterbalanced orders, each lasting 25 minutes: eat with and without TV. After the meal, participants were asked to estimate how much food they had just consumed, using a visible tablespoon as a reference for 'one unit' (which was equivalent to 1.5oz of food).	Food intake (oz) between the TV and no-TV conditions Accuracy of food estimation, determined as estimated units minus actual units consumed 16oz macaroni and cheese provided	There was no difference in amount of food consumed between the TV and no-TV condition (2.5%, p>0.05). Unit estimation error was greater ($T_{0.05}$ 2.2, p=0.022) in the TV vs no-TV condition.
Ogden 2013 [27] Quality = fair	Between subjects UK	Females aged ≥18 who had a manual driving license, n=81 (100%) Mean age 22 (5.18), range 18-40	Aim: to compare the impact of different forms of distraction on eating behavior Protocol: participants were randomized to one of four conditions, each lasting 7 minutes: driving (simulation with a manual transmission), TV (*Friends*), social interaction (conversing with one of the researchers), or sitting alone. 100g of hula hoops (potato snacks) were available in each condition.	Grams of hula hoops consumed 100g were provided (~520kcal)	Those in the TV condition consumed more (28.61 (24.44) g) than those in the social (14.16 (12.33)g, p<0.01) and driving (14.02 (10.34)g, p<0.05) conditions, but no significant difference was seen compared to the alone condition (18.21 (20.9)g, p=0.06).
Ogden 2017 [32] Quality = fair	Between subjects UK	Females from the university, n=60 (100%) Mean age 24 (3.3)	Aim: to compare the impact of distraction and dietary restraint on food intake during a subsequent taste test Protocol: participants were randomized to one of three conditions, each lasting 5 minutes: watching TV, walking (along the corridor), and social interaction (talking with another research participant). During each condition, participants were told to consume a cereal bar. After the condition, participants took part in a taste test lasting 7 minutes during which they could consume as much	Mass and energy (kcal) of food consumed during the taste test 150g of each of the foods were provided	There was no main effect of condition on total mass or total calories consumed during the taste test (values not shown).

			of the foods (M&Ms, carrot sticks, grapes, and hula hoops) as they wished.		
Rosenthal 2017 [26] Quality = good	Within subjects USA	Young adults aged 18 to 35 with BMI between 18.5 and 24.9kg/m^2 n=20 (85%) Mean age 22.3 (3.7)	Aim: to test the independent and interactive effects of TV and portion size on food intake Protocol: participants took part in four randomized conditions one week apart: small portion with and without TV and large portion with and without TV. In the 'no TV' condition, participants ate quietly with no other activities for 30 minutes; the TV condition was a 30-minute episode of *Scandal*.	Food intake (grams and kcals) Small portion condition provided 1083 kcal (650 grams) of macaroni and cheese plus salad with dressing; the large portion provided 2166 kcal (1300 grams) of the same foods	Only a main effect of portion size was found for grams and kcals consumed. No main effect for TV was found for food intake ($F(1,16)=0.39$, $p=0.54$, $d=0.11$) and $F(1,16)=1.10$, $p=0.31$, $d=0.17$)) measured as grams and kcals, respectively. No interactions for TV and portion size were found (>0.05).
Wonderlich-Tierney 2013 [35] Quality = fair	Between subjects USA	Undergraduates, n=83 (52%) Mean age 19.6 (3.5)	Aim: to examine the impact of TV advertisements on food intake Protocol: participants were randomized to one of three groups, each lasting 60 minutes: TV with food advertisements, TV with non-food advertisements, and TV with no advertisements. A jar of cookies was available throughout each condition.	Number of cookies eaten 26 cookies were available in each condition	There was no effect of condition on number of cookies eaten ($F(2,69)=1.50$, $p=0.23$), $R^2=0.04$

Summary of characteristics, measurements, and findings of included studies examining characteristics of sitting or mood while watching TV and sleep after watching TV

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
Characteristics of sitting							
Creasy 2016 [43] Quality = good	Experimental (between subjects) USA	Recruited at university (convenience)	Adults aged 18 to 40 with BMI between 20-35, n=74 (58%)* Mean age 24.2 (3.6) *In full study. Only those whose baseline conditions were sitting while watching TV or sitting at the computer (n=18 in each condition) are described here	Lab-based exposure (15 minutes) *Experimental aim: to examine the cumulative energy expenditure of various combinations of sitting, standing, and walking*	N/A	Energy expenditure (kcal per 15min, indirect calorimetry)	TV EE (18.66 (4.01) kcal/15min) not significantly* different from computer EE (19.63 (4.37) kcal/15min) **Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Mansoubi 2015 [45] Quality = good	Experimental (within subjects) UK	Recruited at university and in community in 2x2 format to obtain equal numbers of men/women and healthy weight/obese participants (convenience)	Adults aged ≥18, 47% of whom were obese, n=51 (51%) Overall mean age not reported	Lab-based exposure (10 minutes) *Experimental aim: to measure the energy expenditure of sedentary behaviours in normal weight and obese participants*	N/A	Energy expenditure (METs, indirect calorimetry)	TV EE (METs) (1.33 (0.24)) was significantly* lower than typing (1.45 (0.32)), playing a handheld computer game (1.41 (0.28)), and playing a screen-based computer game (2.06 (0.50)) * Based on confidence intervals crossing zero in our calculation of the standardized mean change between the conditions; the paper itself did not test this (see Additional file 1)
Newton 2013 [44] Quality = fair	Experimental (within subjects) USA	Recruited in community (convenience)	African American adults aged ≥18, n=25 (60%)	Lab-based exposure (30 minutes) *Experimental aim: to examine the energy*	N/A	Energy expenditure (METs, whole-room calorimetry)	TV EE (METs) (1.03 (0.15)) was not significantly different from EE of typing (1.06 (0.15)), or reading at a desk (1.04 (0.20))

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			Mean age 38.2 (11.4), range 20-56	*expenditure of common sedentary behaviours*			
van der Berg 2019 [46] Quality = fair	Experimental (within subjects) Netherlands	Recruited from the university (convenience)	Adults aged 18 to 48 without mobility limitations, n=18 (44%) Mean age 27.3 (6.5)	Lab-based exposure (7 minutes) seated in a chair without armrests while watching TV on a computer screen *Experimental aim: to identify whether 'dynamic sitting (e.g., fidgeting) can be identified from triaxial accelerometry counts*	N/A	Accelerometer vector magnitude* while watching TV compared to natural sitting (allowed to move freely), imposed sitting (motionless), imposed fidgeting (told to move the upper body side to side and back to front), while reading a newspaper, while writing a letter, while playing cards, while typing on a computer, and while preparing food while seated *Paper reported mean acceleration in the vertical, anteroposterior, and mediolateral planes separately; we converted this to mean vector magnitude by taking the square root of the summed squares of all three axes	Accelerometer vector magnitude was significantly* higher while watching TV compared to natural and 'motionless' sitting, reading, writing, and typing; it was significantly lower while watching TV compared to imposed fidgeting. There was no difference in vector magnitude between TV and playing cards or between TV and preparing food while seated. **Based on confidence intervals crossing zero in our calculation of the standardized mean difference between the two groups; the paper itself did not test this (see Additional file 1)*
Mood							

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Goodwin 2005 [48] Quality = fair	Cross-sectional USA	Convenience sampling	Older adults, n=69 (72%) Mean age 72.0 (6.4)	TV was reported on a paper diary when it was the current activity in response to pager prompts which came 6 times a day for 7 days between 8am and 8pm	N/A NB: affect while watching TV was compared to affect while engaged in other leisure activities (e.g., hobbies, reading), maintenance activities (e.g., cleaning, meal preparation), and productive activities (e.g., work or volunteering) which were reported in response to the pager prompts	Positive and negative affect 'in the moment' when prompted by the pager (i.e., while watching TV if that was the current activity). Mood descriptors were rated on a 5-point Likert scale; positive affect descriptors included things like 'enthusiastic', and negative affect descriptors included things like 'upset'	Positive affect was significantly lower while watching TV (mean 26.07 (SD 9.27)) compared to while doing other leisure activities (28.39 (9.35)) and productive activities (29.78 (9.36)), but was not significantly different compared to while doing maintenance activities (26.19 (9.65)) (see Figure 7). Negative affect while watching TV (10.871 (2.93)) did not significantly differ compared to while doing other leisure activities (10.873 (2.86)), during productive activities (11.09 (2.70)), or during maintenance activities (10.86 (2.74)).
Kuykendall 2020 [47] Quality = good	Cross-sectional USA	Recruited adults who had completed at least 500 assignments on Amazon Mechanical Turk with ≥96% approval rating	Full-time working adults *Study 1a:* n=264 (50%) Mean age 38.7 *Study 1b:* n=538 (50%) Mean age 37.2	*Study 1a & 1b:* Day reconstruction in which all activity episodes from the prior day were reported	N/A	*Study 1a:* For each leisure activity reported in the day reconstruction, the extent to which participants felt the following on a scale of 1 (not at all) to 11 (extremely) was reported: detached, relaxed, stressed,	*Study 1a:* Levels of relaxation were higher when watching TV than when engaging in physical activities; relaxation was similar while watching TV compared to other leisure activities. Levels of satisfaction and happiness were higher during all other leisure activities compared to while watching TV (except for physical activity, for which happiness was

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						autonomy, meaning, mastery, affiliation, happy, sad, satisfied. *Study 1b:* For each leisure activity reported in the day reconstruction, the extent to which participants felt the following: positive activated (energetic, alert, vigorous); negative activated (anxious, jittery, nervous); negative deactivated (bored, tired, dull); or positive deactivated (peaceful, at ease, serene)	similar to TV). Feelings of sadness did not significantly differ while watching TV versus during other leisure activities. *Study 1b:* Positive activated affect was lowest while watching TV compared to all other activities; positive deactivated affect was lower while watching TV than some (but not all) activities. Negative activated affect was lower while watching TV compared to during active, social, and cognitively stimulating activities. Negative deactivated affect was higher while watching TV than during all other activities.
Sleep							
Asaoka 2007 [50] Quality = fair	Intervention Japan	Voluntary sample recruited from pool of previous research participants (university sample) and employment service center (elderly sample)	University students (n=8, 62.5%) and elderly adults (n=8, 62.5%) Mean age for university students 19.9 (0.8), range 19-21; for elderly adults 73.6 (2.7), range 70-78	Logged TV time in daily activity diary (used only to confirm the efficacy of the intervention for changing TV time, not used as a predictor variable)	None	Rising time, bedtime, total sleep time, nap total sleep time, and daytime total sleep time (between 8am and 8pm) all derived from self-reported sleep logs	Compared to the control week, university students went to bed significantly earlier ($p<0.05$), slept for significantly longer in total ($p<0.05$) and during the daytime ($p<0.05$) when TV was limited to 30 minutes per day; there were no significant changes in rising time, bedtime, or total sleep time among the elderly ($p>0.05$).
Komada 2015 [49] Quality = fair	Experimental (between and within subjects)	Convenience sample	Young Japanese men, n=13 (0%)	Lab-based exposure (107 minutes) on three night-time occasions.	N/A	Change in salivary melatonin and cortisol levels (taken both	The increase in melatonin levels was larger following the half-blue light exposure compared to

First author, year Quality assessment	**Study design, country, and name (if applicable)**	**Recruitment strategy**	**Target sample Sample size (% female) Mean (SD) age or age range**	**TV measurement details with operationalization used in analyses**	**Covariates included in model**	**Outcome measurement(s)**	**Finding for the association between TV and outcome measurement(s)**
	Japan		Mean age 22.7 (0.9), range 21-24	Participants were randomized to normal luminance or high luminance display groups. Regardless of group, all participants were exposed to both normal blue light and half blue light as well as a baseline (control) condition (random order).		before and after the 107-minute exposure)	the normal blue light exposure ($p<0.05$); there was no effect of luminance or luminance*blue light interaction on melatonin levels. There were no associations between blue light level, luminance, or their interaction on cortisol levels.